

- 5 A particle  $P$  is projected with speed  $u \text{ m s}^{-1}$  at an angle  $\theta$  above the horizontal from a point  $O$  on a horizontal plane and moves freely under gravity. During its flight  $P$  passes through the point which is a horizontal distance  $3a$  from  $O$  and a vertical distance  $\frac{3}{8}a$  above the horizontal plane. It is given that  $\tan \theta = \frac{1}{3}$ .

(a) Show that  $u^2 = 8ag$ . [2]

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A particle  $Q$  is projected with speed  $V \text{ m s}^{-1}$  at an angle  $\alpha$  above the horizontal from  $O$  at the instant when  $P$  is at its highest point. Particles  $P$  and  $Q$  both land at the same point on the horizontal plane at the same time.

(b) Find  $V$  in terms of  $a$  and  $g$ . [7]

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